Ordinary categorical Langlands correspondence for modular curves

Connections with Iwasawa theory

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Elliptic modular forms and their Galois representations: p-ordinary cuspidal case

p: prime number ≥ 5. $G_p = \mathrm{Gal}(\overline{\mathbf{Q}}_p/\mathbf{Q}_p) \subset G_{\mathbf{Q}} = \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}) \twoheadrightarrow G_{\mathbf{Q},\mathcal{S}}$ cut out by maximal algebraic extension of $\mathbf{Q}$ unramified outside $\mathcal{S} = \{\ell \mid Np\infty\}$, $p \nmid N\phi(N)$. Level $\Gamma_1(Np^r)$.

Hecke eigenform $f = \sum_{n\geq 0} a_n(f)\, q^n$

p-ordinary: $v_p\big(a_p(f)\big) = 0$

$\longleftrightarrow$

$\rho_f : G_{\mathbf{Q}} \longrightarrow \mathrm{GL}_2(\overline{\mathbf{Q}}_p)$

$\mathrm{Tr}\ \rho_f(\mathrm{Fr}_\ell) = a_\ell(f)$ for almost all primes ℓ

p-ordinary: $\rho_f\big|_{G_p} \simeq \left(\begin{smallmatrix} \mathrm{un} & 0 \\ * & * \end{smallmatrix}\right)$, $\mathrm{un}(\mathrm{Fr}_p) = a_p(f)$

Hida theory: eigensystems interpolable over $\Lambda := \mathbf{Z}_p[\![\mathbf{Z}_p^\times]\!]$:

Weight $k \geq 2$, character $\chi : (\mathbf{Z}/p^r\mathbf{Z})^\times \to \overline{\mathbf{Z}}_p^\times \rightsquigarrow \phi_{(k,\chi)} : \Lambda \twoheadrightarrow \mathcal{O}_r \subset \overline{\mathbf{Z}}_p$, $[z] \mapsto z^{k-1}\chi(z)$.

$\rightsquigarrow \mathbf{T}^\circ$: cuspidal ordinary Hecke algebra, finite and flat over Λ. $\quad \mathbf{T}^\circ \otimes_{\Lambda,(k,\chi)} \mathcal{O}_r \cong \mathbf{T}^\circ_k(\chi)^{\mathrm{ord}}$.

$\rightsquigarrow S$: Λ-adic ordinary cusp forms, fin. flat over Λ. $\quad S \cong \mathrm{Hom}_\Lambda(\mathbf{T}^\circ, \Lambda) =: \mathbf{T}^{\circ\vee}$.

Eichler–Shimura isomorphism (Ohta): $H^1(X) := \varprojlim_r H^1_{\text{ét}}(X_1(Np^r), \mathbf{Z}_p(1))^{\mathrm{ord}} \cong \mathbf{T}^{\circ\vee} \oplus \mathbf{T}^\circ$

Fact: $\mathrm{End}_{\mathbf{T}^\circ}(\mathbf{T}^{\circ\vee}) \cong \mathbf{T}^\circ$. So char. poly. of Fr_ℓ on $H^1(X)$ is $x^2 - T_\ell x + [\ell] \in \mathbf{T}^\circ[x]$.

Classical $R = \mathbf{T}$ realized by $H^1(X)$ under genericity conditions

$\mathbf{T}[G_{\mathbf{Q},\mathcal{S}}] \ni \mathrm{Fr}_\ell \curvearrowright H^1(X)$ with $\mathrm{char}(\mathrm{Fr}_\ell, x) = x^2 - T_\ell x + [\ell]$ specializes along $\mathbf{T}^\circ \xrightarrow{f} \overline{\mathbf{Q}}_p$ as

$$\rho_f := H^1(X) \otimes_{\mathbf{T}^\circ, f} \overline{\mathbf{Q}}_p \quad \text{with } \mathrm{char}(\mathrm{Fr}_\ell, x) = x^2 - a_\ell(f)x + \ell^{k-1}\chi(\ell).$$

Modularity lifting: $R_{\bar\tau}^{(\mathrm{ord})}$ is universal ordinary deformation ring of irred. $\bar\tau : G_{\mathbf{Q},\mathcal{S}} \to \mathrm{GL}_2(\overline{\mathbf{F}}_p)$.

Theorem (Wiles, particular case)

Isomorphism $R_{\bar\tau} \cong \mathbf{T}^\circ_{\bar\tau}$ of complete intersection (CI) rings, provided that

- *$\bar\tau$ is p-distinguished + more: $(\bar\tau|_{G_p})^{ss} \simeq \bar\chi_1 \oplus \bar\chi_2$ with $\bar\chi_1\bar\chi_2^{-1} \neq 1, \omega^{\pm 1}$*
- *$\bar\tau$ satisfies a Taylor–Wiles condition and has Artin conductor N ("minimal level")*

$\mathbf{T}^\circ_{\bar\tau}$ is CI $\implies$ $\mathbf{T}^\circ_{\bar\tau} \cong \mathbf{T}^{\circ\vee}_{\bar\tau}$ (i.e. $\mathbf{T}^\circ$ is Gorenstein) $\iff$ $H^1(X)_{\bar\tau} \cong (\mathbf{T}^\circ_{\bar\tau})^{\oplus 2}$.

$\rightsquigarrow$ "$R_{\bar\tau} \twoheadrightarrow \mathbf{T}^\circ_{\bar\tau}$ is an isomorphism" is realized by $H^1(X)$.

Convention: these assumptions remain in force for reducible $\bar\tau$ (except Taylor–Wiles).

p-ordinary cohomology of the open modular curve

Let $\mathbf{T}$ denote the ordinary Hecke algebra, Λ-flat and finite. Likewise, let M be Λ-adic ordinary modular forms. Genericity of $\bar{\tau} = \bar{\chi}_1 \oplus \bar{\chi}_2 \Rightarrow M \cong \mathrm{Hom}_\Lambda(\mathbf{T}, \Lambda) = \mathbf{T}^\vee$. Let $\chi =$ lift of $\bar{\chi}_1 \bar{\chi}_2^{-1}$.

$\mathbf{T} \twoheadrightarrow \mathbf{T}^\circ$ is $(-)^\vee$-dual to SES $0 \to S \hookrightarrow M \xrightarrow{a_0} \Lambda \to 0$.

Cohomology of open modular curve: $H^1(Y) := H^1_{\text{ét}}(Y_1(Np^\infty), \mathbf{Z}_p(1))^{\mathrm{ord}}$ is Λ-flat $\mathbf{T}[G_\mathbf{Q}]$-mod., Poincaré dual to $H^1_c(Y)$ via $\mathrm{Hom}_\Lambda(-, \Lambda^\#(1))$.

Comparison of Ohta's Eichler–Shimura results:

$$\begin{array}{ccccccc} & & H^1_c(Y) & \twoheadrightarrow & H^1(X) & \hookrightarrow & H^1(Y) \\ & & \wr| & & \wr| & & \wr| \\ \mathbf{T}^{\circ\vee} \oplus \mathbf{T} & \xrightarrow{\sim} & S \oplus M^\vee & \twoheadrightarrow & S \oplus S^\vee & \hookrightarrow & M \oplus S^\vee \xrightarrow{\sim} \mathbf{T}^\vee \oplus \mathbf{T}^\circ. \end{array}$$

OTOH, there are Eisenstein series $\Lambda \xrightarrow{\mathrm{Eis}} M$, dual to the Eisenstein quotient $\mathbf{T} \twoheadrightarrow \mathbf{T}/I \cong \Lambda$.

The composition $\Lambda \xrightarrow{\mathrm{Eis}} M \xrightarrow{a_0} \Lambda$ given by ξ: p-adic Dirichlet L-func. On branch $\bar{\tau}$, $\xi \rightsquigarrow \xi_\chi \in \Lambda_\chi$.

Facts: $\xi_\chi \in \Lambda_\chi^\times \iff S_{\bar{\tau}} = 0$. Moreover, $\mathbf{T}^\circ \otimes_\mathbf{T} \mathbf{T}/I \cong \mathbf{T}^\circ_{\bar{\tau}}/I^\circ \cong \Lambda_\chi/\xi_\chi$ (Emerton, 1999).

Iwasawa main conjecture $\leftrightarrow$ p-ordinary residually Eisenstein modular forms

The approach to the IMC of Mazur–Wiles, in modern terms (Ohta, Sharifi, Ochiai's book):

Step 1: The congruence quotient between cuspidal and Eisenstein

This is $\mathbf{T}^\circ \otimes_{\mathbf{T}} \mathbf{T}/I \cong \mathbf{T}^\circ/I^\circ \cong \Lambda/\xi$.

Step 2: Reducibility modulo I° of $G_{\mathbf{Q}}$-action produces unramified extension fields

Consider $G_{\mathbf{Q}}$-action on $H^1(X) \otimes_\Lambda Q(\Lambda) \simeq Q(\mathbf{T}^\circ)^{\oplus 2} \rightsquigarrow$ let image of $\Lambda[G_{\mathbf{Q}}]$ be $\left(\begin{smallmatrix} \mathbf{T}^\circ & B' \\ C' & \mathbf{T}^\circ \end{smallmatrix}\right) \subset \mathrm{End}$ and $G_{\mathbf{Q}_p}$ image is lower-triangular, hence get p-unramified 1-cocycle by reduction modulo I°

$$G_{\mathbf{Q}} \twoheadrightarrow B'/I^\circ B' \quad \Rightarrow \quad \text{pre-}\Upsilon : X^- \twoheadrightarrow B'/I^\circ B'$$

B' is a faithful $\mathbf{T}^\circ$-module $\Rightarrow$ Iwasawa ideal class group $X^- = \varprojlim_r \mathrm{Cl}(\mathbf{Q}(\zeta_{Np^r}))[p^\infty]^-$ has Fitting ideal $\subseteq \xi\Lambda \Leftrightarrow$ IMC, i.e., $\mathrm{char}_\Lambda(X^-) = \xi\Lambda$.

Sharifi's conjecture:* canonical lattice $H^1(X) \cong S \oplus \mathbf{T}^\circ \pmod{I^\circ}$ yields $\boxed{\Upsilon : X^- \xrightarrow{\sim} S/I^\circ S.}$ using b-coord. $S = \mathrm{Hom}_{\mathbf{T}^\circ}(\mathbf{T}^\circ, S) \subset \mathrm{End}_{\mathbf{T}^\circ}(H^1(X))$.

(*approx. half of Sharifi's conj. (2011))

The p-ordinary modular curve case of Emerton and Zhu's categorical Langlands conjecture integrates

residually irreducible $R = \mathbf{T}$

$$G_{\mathbf{Q}} \curvearrowright \quad R_{\bar{\tau}} \oplus R_{\bar{\tau}} \;\cong\; H^1(X)_{\bar{\tau}} \;\cong\; S_{\bar{\tau}} \oplus \mathbf{T}^{\circ}_{\bar{\tau}} \;\cong\; \mathbf{T}^{\circ}_{\bar{\tau}} \oplus \mathbf{T}^{\circ}_{\bar{\tau}}$$

with residually reducible $R^{\mathrm{ps}} = \mathbf{T}$

along with Mazur–Wiles (IMC) and Sharifi's conjecture:

$$G_{\mathbf{Q}} \curvearrowright \quad H^1(X)_{\bar{\tau}}/I \cong S_{\bar{\tau}}/IS_{\bar{\tau}} \oplus \mathbf{T}^{\circ}_{\bar{\tau}}/I\mathbf{T}^{\circ}_{\bar{\tau}} \quad \rightsquigarrow \quad \Upsilon : X^{-} \xrightarrow{\sim} S_{\bar{\tau}}/I^{\circ}S_{\bar{\tau}}.$$

Stack of Galois representations over moduli of pseudorepresentations

Excerpt from Mark Kisin's *Deformations of Galois Reps.* (Clay Summer School, Hawaii, 2009).

(3.2) Now fix a pseudo-representation $\bar{\tau} : G \to \mathbb{F}$ We want to construct some representable subgroupoids of $\mathrm{Rep}_{\bar{\tau}} \to D_{\bar{\tau}}$.

Suppose that for $i = 1, \ldots, s$ there are pairwise distinct absolutely irreducible representations $\bar{\rho}_i : G \to \mathrm{GL}_{d_i}(\mathbb{F})$ of G, such that $\bar{\tau} = \sum_{i=1}^{s} \bar{\rho}_i$. Let $\mathrm{Rep}'_{\bar{\tau}} \subset \mathrm{Rep}_{\bar{\tau}}$ be the full subgroupoid of $\mathrm{Rep}_{\bar{\tau}}$ such that $\mathrm{Rep}'_{\bar{\tau}}(A, B)$ consists of the objects V_B in $\mathrm{Rep}_{\bar{\tau}}(A, B)$ such that locally in the Zariski topology of $\mathrm{Spec}\, B$,

$$V_B \otimes_B B/\mathfrak{m}_A B \sim \begin{pmatrix} \bar{\rho}_1 & * & \cdots & \cdots \\ & \bar{\rho}_2 & * & \cdots \\ & & \cdots & * \\ & & & \bar{\rho}_s \end{pmatrix} \tag{3.2.1}$$

and is required to satisfy the following condition: Let

$$0 = L_0 \subset L_1 \subset \ldots L_s = V_B \otimes_B B/\mathfrak{m}_A B$$

be the filtration with $L_i/L_{i-1} \sim \bar{\rho}_i$. Then the extension class of L_i as an extension of $\bar{\rho}_i$ by L_{i-1} is non-trivial.

Note that the isomorphism in (3.2.1) is uniquely determined as our conditions imply that the representation on the right has no non-trivial automorphisms.

Theorem (3.2.2). *The groupoid* $\mathrm{Rep}'_{\bar{\tau}} \to D_{\bar{\tau}}$ *is representable by a proper formal scheme over* $\mathrm{Spf}\, R_{\bar{\tau}}$.

Remark (3.2.7) I expect that the representing formal scheme in (3.2.2) is projective and therefore arises from a *scheme* of finite type over $\mathrm{Spec}\, R_{\bar{\tau}}$.

Distinguished residually reducible ordinary case for modular curves

Theorem (WE; arbitrary dimension of representations)

$\psi : \mathrm{Rep}_{\bar\tau} \to \mathrm{Spec}\, R^{\mathrm{ps}}_{\bar\tau}$ is finite type and universally closed. If $\bar\tau$ has s distinct irred. summands, then $\mathrm{Rep}_{\bar\tau}$ is presentable as a quotient $[\mathrm{Spec}\, W_{\bar\tau}/\mathbf{G}_m^{s-1}]$ and ψ is a GIT quotient map.

We go back to 2-dimensional reps: so $s - 1 = 1$ when $\bar\tau = \bar\chi_1 \oplus \bar\chi_2$, $\bar\chi_1 \neq \bar\chi_2$.

$W_{\bar\tau} = W$ is a $\mathbf{Z}$-graded ring, which has the same info as the associated generalized matrix algebra (GMA) $E_{\bar\tau}$,

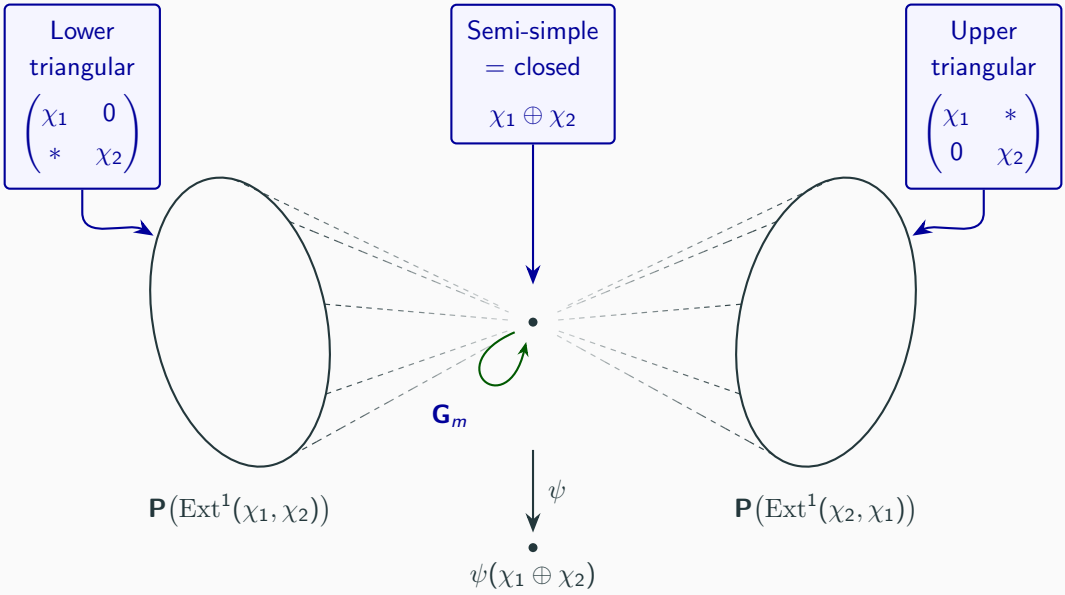
$$E_{\bar\tau} = \begin{pmatrix} R^{\mathrm{ps}}_{\bar\tau} & W_2 \\ W_{-2} & R^{\mathrm{ps}}_{\bar\tau} \end{pmatrix} := R^{\mathrm{ps}}_{\bar\tau}[G_{\mathbf{Q},S}]/(\text{Cayley–Hamilton}).$$

$\mathrm{Rep}_{\bar\tau}$ is presented as the moduli scheme of standard matrix idempotent-preserving $R^{\mathrm{ps}}_{\bar\tau}$-maps from $E_{\bar\tau}$ to $M_{2\times 2}$ (Bellaïche–Chenevier), quotient by the adjoint action of the torus of ~~PGL_2~~ $\mathrm{SL}_2 \curvearrowright M_{2\times 2}$.

$$W_\bullet = \cdots \oplus W_{-4} \oplus W_{-2} \oplus (W_0 = R^{\mathrm{ps}}) \oplus W_2 \oplus W_4 \oplus \cdots$$

Kisin's $\mathrm{Rep}'_{\bar\tau}$ is Proj of $W_{\geq 0}$.

Fiber of ψ over a reducible pseudorepresentation



Emerton and Zhu's conjecture

9.3.1. *Cohomology of Shimura varieties.* Suppose now that $F = \mathbf{Q}$, and that the reductive group G admits a Shimura datum. We let F (which is now freed up as a piece of notation!) denote the reflex field of this Shimura datum, and let $\mu : (\mathbf{G}_m)_{/F} \to G_{/F}$ denote the associated Hodge cocharacter (well-defined up to conjugacy). In the definition of ${}^cG$, we choose the extension $\widetilde{F}$ of $\mathbf{Q}$ large enough that it contains F, and then write

$$ {}^cG_F := \widehat{G} \rtimes \left(\mathbf{G}_m \times \mathrm{Gal}(\widetilde{F}/F)\right) \subseteq \widehat{G} \rtimes \left(\mathbf{G}_m \times \mathrm{Gal}(\widetilde{F}/\mathbf{Q})\right) =: {}^cG. $$

The Hodge cocharacter μ, thought of as a weight of $\widehat{G}$, gives rise to an irreducible (highest weight) representation V_μ of $\widehat{G}$, which naturally extends to a representation of ${}^cG_F$. For any set S of primes (containing p and ∞ as usual), the moduli stack $\mathcal{X}_{G,\mathbf{Q},S}$ parameterizes a universal family of (derived) C-homomorphisms $\mathrm{Gal}_{\mathbf{Q}} \to {}^cG$, which we can restrict to Gal_F so as to obtain a family of representations $\mathrm{Gal}_F \to {}^cG_F$. Composing this family with the representation V_μ of ${}^cG_F$, we obtain a vector bundle $\mathcal{V}_\mu$ over $\mathcal{X}_{G,\mathbf{Q},S}$, which is endowed with an action of Gal_F.

Choose a level subgroup K_f, i.e. a compact open subgroup of $G(\mathbf{A})$, which we suppose may be factored as $K_f = \prod_v K_v$, with K_v compact open in $G(\mathbf{Q}_v)$ for each finite place v. We choose (as we may) S large enough that G is unramified outside S, and so that there is an extension $\mathcal{G}$ of G to a reductive group scheme over $\mathbf{Z}[1/S]$ such that $K_v = \mathcal{G}(\mathbf{Z}_v)$ for $v \notin S$. The locally symmetric quotient $Y(K_f) := G(\mathbf{Q})\backslash\left(X \times G(\mathbf{A}_f)\right)/K_f$ is a Shimura variety.

Excerpt from Emerton–Gee–Hellmann *Intro. categorical p-adic Langlands program* (IHÉS summer school, 2022)

For each finite prime $v \in S$, we let W_v be a smooth representation of K_v on a finite torsion $\mathcal{O}$-module. We let $W := \bigotimes_{v\in S} W_v$ denote the corresponding representation of K_f (which acts through its projection onto $\prod_{v\in S} K_v$; the tensor product is taken over $\mathcal{O}$, of course), and we let $\mathcal{W}$ denote the associated local system $\mathcal{W} := G(\mathbf{Q})\backslash\left(X \times G(\mathbf{A}_f) \times W\right)/K_f$ over $Y(K_f)$. For each v, the compact induction $c\text{-}\mathrm{Ind}_{K_v}^{G(\mathbf{Q}_v)} W_v$ is a smooth $G(\mathbf{Q}_v)$-representation. Applying the functors of Conjectures 6.1.15 and 8.3.1, we obtain (conjecturally!) a coherent sheaf on $\mathcal{X}_{G,\mathbf{Q}_v}$, which we denote simply by $\mathfrak{A}_v$. We may form the exterior tensor product $\mathfrak{A} := \boxtimes_{v\in S_f} \mathfrak{A}_v$ on $\prod_{v\in S_f} \mathcal{X}_{G,\mathbf{Q}_v}$.

As above, let $f : \mathcal{X}^{\mathrm{odd}}_{G,\mathbf{Q},S} \to \prod_{v\in S} \mathcal{X}_{G,\mathbf{Q}_v}$ denote the global-to-local restriction morphism. We then have the following conjectural formula for the cohomology of the Shimura variety.

Conjecture 9.3.2. *Let d be the complex dimension of $Y(K_f)$. Then there is an isomorphism*

$$ R\Gamma_c\big(Y(K_f), \mathcal{W}\big)[d] \xrightarrow{\sim} R\Gamma\big(\mathcal{X}^{\mathrm{odd}}_{G,\mathbf{Q},S}, f^!\mathfrak{A} \otimes \mathcal{V}_\mu\big) \tag{9.3.3} $$

compatible with the Gal_F- and (derived) Hecke actions on source and target.

Conjecture (Emerton and Zhu, p-ordinary modular curve case of Conj. 9.3.2)

$$ H^1_c(Y)_{\bar\tau} \cong H^0\big(\mathrm{Rep}_{\bar\tau}, f_p^!\mathfrak{A}_p^{\mathrm{ord}} \otimes \mathcal{V}_{\mathrm{std}}\big) \qquad \big(\mathbf{T}[G_{\mathbf{Q}}] \twoheadleftarrow R^{\mathrm{ps}}[G_{\mathbf{Q}}]\big)\text{-equivariantly.} $$

Simplified using trivialization of ramified Hecke operators.

Next: interpret the conjecture using $\mathbf{G}_m$-equivariant structure of $\mathrm{Rep}_{\bar\tau}$.

$\mathbf{G}_m$-equivariant structure on the spectral side of Emerton and Zhu's conjecture

Normalize the adjoint action of $\mathbf{G}_m \subset \mathrm{SL}_2$: the b-coordinate of $M_{2\times 2}$ by $2 \in \mathbf{Z} = X^*(\mathbf{G}_m)$, hence by -2 on the c-coordinate, corresponding to $\mathcal{V}_{\mathrm{std}} \cong \mathcal{O}(\text{-}1) \oplus \mathcal{O}(1) \quad = \begin{pmatrix} \mathcal{O}(\text{-}1) \\ \mathcal{O}(1) \end{pmatrix}$

$$\rightsquigarrow \qquad \mathcal{E}nd(\mathcal{V}_{\mathrm{std}}) \cong \begin{pmatrix} \mathcal{O} & \mathcal{O}(\text{-}2) \\ \mathcal{O}(2) & \mathcal{O} \end{pmatrix} \quad \supset \quad \mathcal{E}nd(\mathcal{V}_{\mathrm{std}})^{\mathbf{G}_m} = E.$$

Recall we normalize "ordinary" ρ so that $\rho|_{G_p} \simeq \begin{pmatrix} \mathrm{un} & 0 \\ * & * \end{pmatrix}$.

Theorem (Johansson–Newton–WE, Dotto–Emerton–Gee)

Given $\bar{\tau} = \bar{\chi}_1 \oplus \bar{\chi}_2$ generic rep. of G_p, the level sheaf $\mathfrak{A}_p$ on $\mathrm{Rep}_{p,\bar{\tau}}$ representing the ordinary condition is the (-1)*-twist structure sheaf of the (closed) ordinary locus, $\mathcal{O}_{\mathrm{Rep}_p^{\mathrm{ord}}}(\text{-}1)$.*

$$\mathrm{Rep}_{\mathbf{G}_m \times \mathbf{G}_m, p} \xleftarrow{p} \mathrm{Rep}_{B,p} \xrightarrow{q} \mathrm{Rep}_p \overset{\text{c.i.}}{\supset} \mathrm{Rep}_p^{\mathrm{ord}}, \qquad \mathfrak{A}_p^{\mathrm{ord}} = q_* p^* \mathfrak{A}_{\mathbf{G}_m \times \mathbf{G}_m, p}(\delta^{\mathrm{ord}}).$$

Presuming that $f : \mathrm{Rep}_{\bar{\tau}} \to \mathrm{Rep}_{\bar{\tau},p}$ is a complete intersection morphism so that $f^! = f^*$:

Conjecture (E–Z(classical), interpreted $\mathbf{G}_m$-equivariantly)

$$H^1_c(Y)_{\bar{\tau}} \cong H^0(\mathrm{Rep}_{\bar{\tau}}, \mathcal{O}^{\mathrm{ord}}(\text{-}1) \otimes \mathcal{V}_{\mathrm{std}}) \cong H^0(\mathrm{Rep}_{\bar{\tau}}^{\mathrm{ord}}, \mathcal{O}(\text{-}2) \oplus \mathcal{O}).$$

Ordinary representations and pseudorepresentations

An intrinsic notion of ordinary representations and pseudorepresentations.

Theorem (Wake–WE (2017))

Under the running hypotheses on $\bar{\tau}$, there is $\psi^{\mathrm{ord}} : \mathrm{Rep}_{\bar{\tau}}^{\mathrm{ord}} \to \mathrm{Spec}\, R_{\bar{\tau}}^{\mathrm{ps,ord}}$ cut out by the ordinary condition $\rho|_{G_p} \simeq \left(\begin{smallmatrix} \mathrm{un} & 0 \\ * & * \end{smallmatrix}\right)$. The moduli stack $\mathrm{Rep}_{\bar{\tau}}^{\mathrm{ord}} = [\mathrm{Spec}\, W_{\bullet\,\bar{\tau}}^{\mathrm{ord}} / \mathbf{G}_m]$ is the $\mathbf{G}_m$-quotient stack of the moduli $\mathrm{Spec}\, W_{\bullet\,\bar{\tau}}^{\mathrm{ord}}$ of idempotent preserving $M_{2\times 2}$-representations of a universal ordinary GMA over $R_{\bar{\tau}}^{\mathrm{ps,ord}}$,

$$E_{\bar{\tau}}^{\mathrm{ord}} = \begin{pmatrix} R_{\bar{\tau}}^{\mathrm{ps,ord}} & W_{2\,\bar{\tau}}^{\mathrm{ord}} \\ W_{\text{-}2\,\bar{\tau}}^{\mathrm{ord}} & R_{\bar{\tau}}^{\mathrm{ps,ord}} \end{pmatrix} = R^{\mathrm{ps}}[G_{\mathbf{Q},S}] \Big/ \begin{pmatrix} \text{C–H relations,} \\ \text{ord. relations and their } \mathrm{Tr} + \det \end{pmatrix}$$

Since we only work with the global ordinary moduli spaces localized at good $\bar{\tau}$ from now on, we drop the decorations $(-)_{\bar{\tau}}^{\mathrm{ord}}$ from the notation.

Conjecture (E–Z(classical), interpreted $\mathbf{G}_m$-equivariantly)

$$H_c^1(Y) \cong H^0(\mathrm{Rep}, \mathcal{O}(\text{-}2) \oplus \mathcal{O}).$$

The meaning of the conjecture

Recall Eichler–Shimura isomorphism $\mathbf{T}^{\circ\vee} \oplus \mathbf{T} \cong S \oplus M^\vee \cong H^1_c(Y)$.

Conjecture (E–Z(classical), ordinary case)

$$S \oplus \mathbf{T} \cong H^1_c(Y) \cong W_2 \oplus R^{\mathrm{ps}} = \textit{right column of } E.$$

Recall $E = \left(\begin{smallmatrix} R^{\mathrm{ps}} & W_2 \\ W_{-2} & R^{\mathrm{ps}} \end{smallmatrix}\right)$. The conjecture is saying:

- $H^1_c(Y)$ is a projective E-module (its right column), the projective envelope of $\bar{\chi}_2$
- Intrinsic definition of cuspidal on $\mathrm{Rep}^{\mathrm{ord}}$: max. rank locus of $H^0(\mathrm{Rep}^{\mathrm{ord}}, \mathcal{O}(\text{-}1) \otimes \mathcal{V}_{\mathrm{std}})$.

Why it's believable:

- Of course we believe $R^{\mathrm{ps}} \cong \mathbf{T}$ (ordinary).
- Check rank at minimal primes of $\mathbf{T}$: while $H^0(\mathcal{O}(2n))$ is rank 1 on the cuspidal components for any n, on the Eisenstein component $\mathbf{T} \twoheadrightarrow \mathbf{T}/I \cong \Lambda$,

$$E/I^{\mathrm{ps}}E \cong \begin{pmatrix} \Lambda & X^- \\ \mathfrak{X}^- & \Lambda \end{pmatrix} = \begin{pmatrix} H^0(\mathcal{O}/I^{\mathrm{ps}}) & H^0(\mathcal{O}(\text{-}2)/I^{\mathrm{ps}}) \\ H^0(\mathcal{O}(2)/I^{\mathrm{ps}}) & H^0(\mathcal{O}/I^{\mathrm{ps}}) \end{pmatrix} \overset{Q(\Lambda)}{\rightsquigarrow} \begin{pmatrix} Q(\Lambda) & 0 \\ Q(\Lambda) & Q(\Lambda) \end{pmatrix}.$$

Recall Ohta's Eichler–Shimura isomorphism $\mathbf{T}^{\circ\vee} \oplus \mathbf{T} \cong S \oplus M^{\vee} \cong H^1_c(Y)$.

E–Z's conjecture (ordinary case): $H^1_c(Y) \cong W_2 \oplus R^{\mathrm{ps}}$ as $R^{\mathrm{ps}}[G_{\mathbf{Q}}]$-modules.

Theorem (WE, in progress)

The E–Z(classical) conjecture of a $\mathbf{T}[G_{\mathbf{Q}}]$-equivariant isomorphism $H^1_c(Y) \cong W_2 \oplus R^{\mathrm{ps}}$ is equivalent to $R^{\mathrm{ps}} \cong \mathbf{T}$ and Sharifi's conjecture that $\Upsilon : X^- \to S/IS$ is an isomorphism.

Proof.

Spec $W_\bullet$ is the moduli functor $A \mapsto \{\text{coord.-preserving } E \to M_{2\times 2}(A)\}$. Recall

$$R^{\mathrm{ps}}[G_{\mathbf{Q},S}] \overset{\text{univ}}{\twoheadrightarrow} E := \begin{pmatrix} R^{\mathrm{ps}} & W_2 \\ W_{-2} & R^{\mathrm{ps}} \end{pmatrix} \overset{(\mathrm{mod}\ I^{\mathrm{ps}})}{\twoheadrightarrow} \begin{pmatrix} \Lambda & X^- \\ \mathfrak{X}^- & \Lambda \end{pmatrix}$$

The natural $R^{\mathrm{ps}}[G_{\mathbf{Q},S}]$-action on $H^0(\mathrm{Rep}^{\mathrm{ord}}, \mathcal{O}(\text{-}1) \otimes \mathcal{V}_{\mathrm{std}}) \cong W_2 \oplus R^{\mathrm{ps}}$ matches the left regular action of E on its right column. We see the E–Z(cl.) conjecture implies $R^{\mathrm{ps}} \cong \mathbf{T}$. Given $R^{\mathrm{ps}} \cong \mathbf{T}$, an isom $\Upsilon : X^- \xrightarrow{\sim} S/IS \Leftrightarrow W_2 \twoheadrightarrow S \Leftrightarrow W_2 \cong S$. □

The conjecture is known when T is complete intersection (CI).

There are cases in which we know that $R^{\mathrm{ps}} \cong \mathbf{T}$ and Υ is an isom. $\bar{\tau} = \bar{\chi}_1 \oplus \bar{\chi}_2$; $\bar{\chi} = \bar{\chi}_1 \bar{\chi}_2^{-1}$.

Theorem (Wake–WE (2017))

If $\mathfrak{X}^-_{\chi^{-1}}$ is Λ_χ-cyclic, then $R^{\mathrm{ps}}_{\bar{\tau}} \cong \mathbf{T}_{\bar{\tau}}$ is an isomorphism of CI rings and Υ_χ is an isomorphism.

Theorem (Wake (2015))

Assume $\mathbf{T}_{\bar{\tau}} \twoheadrightarrow \mathbf{T}_{\bar{\tau}}/I = \Lambda$ is a strict surjection ($\Leftrightarrow \xi_\chi \notin \Lambda_\chi^\times \Leftrightarrow X_\chi^- \neq 0 \Leftrightarrow \mathbf{T}^\circ_{\bar{\tau}} \neq 0$).
If $\mathbf{T}_{\bar{\tau}}$ is Gorenstein, then $\mathfrak{X}^-_{\chi^{-1}}$ is cyclic.

Corollary (Wake, Wake–WE)

Assume $X_\chi^- \neq 0$. There are equivalences $\mathbf{T}_{\bar{\tau}}$ CI $\Leftrightarrow$ $\mathbf{T}_{\bar{\tau}}$ Gorenstein $\Leftrightarrow$ $\mathfrak{X}^-_{\chi^{-1}}$ is cyclic.
When these equivalent conditions are true, $\Upsilon_\chi : X_\chi^- \to S_{\bar{\tau}}/IS_{\bar{\tau}}$ is an isomorphism.

Corollary (Combination of Wake–WE (2017) and WE (in progress))

If $\mathfrak{X}^-_{\chi^{-1}}$ is cyclic, then conjecture E–Z(classical) holds: $H^1_c(Y)_{\bar{\tau}} \cong H^0(\mathrm{Rep}^{\mathrm{ord}}_{\bar{\tau}}, \mathcal{O}(\text{-}1) \otimes \mathcal{V}_{\mathrm{std}})$.

Cyclicity of $\mathfrak{X}^-$ and its link with CI-ness of T

Example (Wake)

It is very often the case that $\mathfrak{X}^-_{\chi^{-1}}$ is $\simeq \Lambda_\chi$, but there are exceptions, e.g. for $p = 5$, $N = 350267$, certain $\bar{\tau}$ where $\chi = \chi^{-1}$.

The role of cyclicity of $\mathfrak{X}^-_{\chi^{-1}}$, i.e. $\mathfrak{X}^-_{\chi^{-1}} \simeq \Lambda_\chi$, to prove $R^{\mathrm{ps}}_{\bar{\tau}} \cong \mathbf{T}_{\bar{\tau}}$.

- We know that $\mathbf{T}^\circ_{\bar{\tau}} \otimes_{\mathbf{T}_{\bar{\tau}}} \mathbf{T}_{\bar{\tau}}/I \cong \Lambda_\chi/\xi_\chi$ and that $\mathrm{char}_{\Lambda_\chi}(X^-_\chi) = \xi_\chi \Lambda_\chi$.
- To run the numerical criterion on $R^{\mathrm{ps}}_{\bar{\tau}} \twoheadrightarrow \mathbf{T}_{\bar{\tau}}$, we need $\mathrm{Fitt}_{\Lambda_\chi}(I^{\mathrm{ps}}/(I^{\mathrm{ps}})^2) \subseteq \xi_\chi \Lambda_\chi$.
- It receives a surjection $\mathfrak{X}^-_{\chi^{-1}} \otimes_{\Lambda_\chi} X^-_\chi \twoheadrightarrow I^{\mathrm{ps}}/(I^{\mathrm{ps}})^2$. IMC + cyclicity of $\mathfrak{X}^-_{\chi^{-1}}$ is enough!

Remark

Wake and I also establish a variant of our 2017 result up to pseudo-isomorphism: in particular, If $\mathfrak{X}^-[1/p]$ is $\Lambda[1/p]$-cyclic, then $R^{\mathrm{ps}}_{\mathfrak{p}} \cong \mathbf{T}_{\mathfrak{p}}$ at all Eisenstein/cuspidal intersection points $\mathfrak{p} \in \mathrm{Spec}\, \mathbf{T}[1/p]$ and $\Upsilon[1/p] : X^-[1/p] \to (S/IS)[1/p]$ is an isomorphism.

Upshot: Same argument shows "$\mathfrak{X}^-[1/p]$ cyclic implies conj. E–Z(classical)" after $- \otimes_{\mathbf{T}} \mathbf{T}_{\mathfrak{p}}$.

The $\Lambda[1/p]$-cyclicity of $\mathfrak{X}^-[1/p]$ is a consequence of Greenberg's pseudo-nullity conjecture.

Happy Birthday Matt and Mark!



